

Midterm Exam 1
CENG 113, SP26
Dr. Drews

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Question 1: Estimate K_0 at 400 K for a liquid-phase equilibrium reaction with $\Delta G_{T_R}^\circ = 5 \text{ kJ mol}^{-1}$, $\Delta H_{T_R}^\circ = -15 \text{ kJ mol}^{-1}$, and $\Delta C_P = 0$.

$$K_0 = \exp(-\Delta G/RT) = \exp(-5000/(8.314 \cdot 298)) = 0.133$$

Question 2: Estimate K_1 given the information in the previous problem.

$$K_1 = \exp(\Delta H/R \cdot (1/298 - 1/T)) = \exp(-15000/8.314 \cdot (1/298 - 1/400)) = 0.214$$

Question 3: What's the elementary rate expression for the reaction $A \rightleftharpoons 2B$?

- (A) $r = kC_A$
- (B) $r = kC_B^2$
- (C) $r = k \left(C_A - \frac{C_B}{K_C} \right)$
- ~~(D) $r = k \left(C_A - \frac{C_B^2}{K_C} \right)$~~
- (E) none of the above

Question 4: Volumetric flow rate is always constant for ideal gases if the system is isothermal and isobaric.

- (A) true
- ~~(B) false~~
- (C) not enough information to decide

Question 5: Enthalpy of reaction is minimized at equilibrium.

- (A) true
- ~~(B) false~~
- (C) not enough information to decide

Question 6: Select the best activity for a pure liquid.

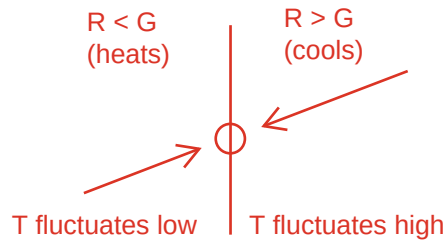
- (A) P_i/P°
- (B) M_i/M_i°
- ~~(C) x_i~~
- (D) 1
- (E) none of the above

Question 7: The equilibrium constant K_C never has units.

- (A) true
- ~~(B) false~~
- (C) not enough information to decide

Question 8: If $R > G$ above a steady-state operating temperature and $R < G$ below, this steady state is best classified as which of the following?

- ☒ (A) stable
- ☐ (B) unstable
- ☐ (C) metastable
- ☐ (D) not enough information to decide



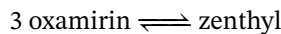
Question 9: If the reaction $A \rightleftharpoons B$ is rewritten as $2 A \rightleftharpoons 2 B$ then $\Delta H_{T_R}^\circ$ will double.

- ☒ (A) true
- ☐ (B) false
- ☐ (C) not enough information to decide

Question 10: Estimate the CSTR reactor volume (in liters) needed to convert 80% of A according to the liquid-phase, elementary reaction $A \longrightarrow \text{products}$ ($k = 6 \text{ s}^{-1}$) assuming a feed of pure A at 2 L s^{-1} .

$$V = n_{A0} \cdot X_a / (-r_A) = n_{A0} \cdot X_a / (k \cdot C_{A0} \cdot (1 - X_a)) = (v/k) \cdot X_a / (1 - X_a) = (2/6) \cdot 0.80 / (1 - 0.80) = 1.333 \text{ L}$$

Question 11: For the liquid-phase equilibrium reaction



determine the temperature T (in K) to achieve 50% conversion of a feed containing pure oxamirin. Thermodynamic data are provided in the table shown, in kJ mol^{-1} .

species	liquid phase		gas phase	
	$\Delta H_{f,i}^\circ$	$\Delta G_{f,i}^\circ$	$\Delta H_{f,i}^\circ$	$\Delta G_{f,i}^\circ$
lumetra	-277	-174	-235	-167
oxamirin	-173	-9	-146	-8
veltrane	-285	-237	-241	-228
zenthyl	-156	26	-123	31

Basis: 6 mol A fed (could choose anything)

Outlet: $n_A = 3$ (50% conv), $n_B = 1$ (stoich), $n_T = n_A + n_B = 3 + 1 = 4$

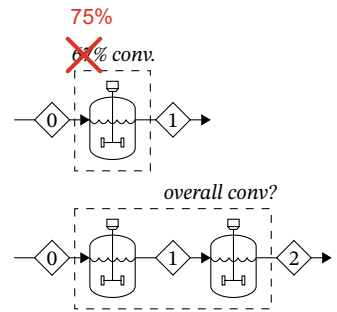
$$\text{prod}(a_i \wedge \nu_i) = x_B / x_A^3 = n_T^2 \cdot n_B / n_A^3 = 4^2 \cdot 1 / (3^3) = 0.593$$

$$K_0 = \exp(-dG/(R \cdot 298)) = 5.124 \text{e-}10$$

$$K_1 = \text{prod}(a_i \wedge \nu_i) / K_0 = 0.593 / 5.124 \text{e-}10 = 1.16 \text{e+}09$$

$$1/T = 1/298 - R/dH \cdot \ln(K_1) \rightarrow T = 348 \text{ K}$$

Question 12: The elementary, liquid-phase scaffloration reaction $A \rightleftharpoons B$ has equilibrium constant $K_C = 6$. Compounds A and B are miscible and of equal density. A feed of pure A achieves 75% conversion in a single isothermal CSTR. Determine the *overall* conversion of A if the product from the first reactor is sent directly to a second reactor with the same operating conditions as the first (temperature, volume, etc.).



In both cases $r = k^*(C_A - C_B/K_C) = k^*C_{A0}^*(1 - X_A - (\theta_B + X_A)/K_C)$

Assume a basis of 4 mol/s A fed (not required but helps with math)

FIRST REACTOR

Sp	nu	ni0	ni
A	-1	4	1
B	1	0	3

$$V = n_{A0}X_A/(-r_A) = n_{A0}X_A/(k^*C_{A0}^*(1 - X_A - (0 + X_A)/K_C))$$

from which

$$\tau^*k = X_A/(1 - X_A - X_A/K_C) = 6 = \text{constant (same temp, same } v_0, \text{ latter b/c same density)}$$

SECOND REACTOR

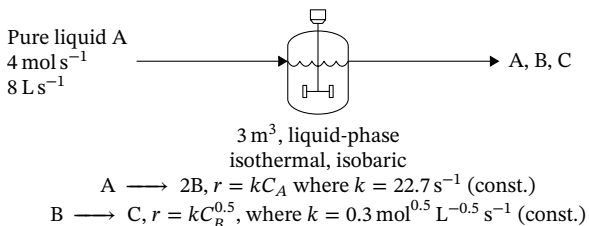
Sp	nu	ni0	ni
A	-1	1	$1^*(1 - X_A)$
B	1	3	$1^*(3 + X_A)$

(<--- this is n_{A2})

$$\tau^*k = X_A/(1 - X_A - (3 + X_A)/6) = 6 \text{ from which } X_A = 0.375$$

$$\text{Overall conversion: } X_{A,ov} = 1 - n_{A2}/n_{A0} = 1 - 1^*(1 - 0.375)/4 = 0.844 = 84.4\%$$

Question 13: On the back of your answer sheet write a MATLAB script which uses fsolve to determine the outlet molar flow rate of all species for the reactor shown below. Use the general layout of a script with a local function described in class (other approaches, such as symbolic math or anonymous functions, will not receive credit).



```

1 % [nA, nB, nC] in mol/s
2 n = fsolve(@myRFF, [2 2 2]); % guess can be anything
3
4 function out = myRFF(n)
5 % n = [nA, nB, nC] in mol/s
6 nA = n(1); nB = n(2); nC = n(3);
7 v = 8; % L/s, v = v0 since liq
8 V = 3000; % L
9 r1 = 22.7*nA/v; % mol/L-s
10 r2 = 0.3*(nB/v).^(0.5); % mol/L-s
11
12 % show net rates for clarity
13 out(1,1) = 4 + (-1*r1 + 0*r2)*V - nA;
14 out(1,2) = 0 + ( 2*r1 - 1*r2)*V - nB;
15 out(1,3) = 0 + ( 0*r1 + 1*r2)*V - nC;
16 end

```